Chapter 3

# Electricity Around Us

Electricity makes our lives easier. At home, it runs fans, lights, televisions, refrigerators and mobile phone chargers. Most of the electricity we use comes from big power stations and reaches our homes through wires. Small machines like torches, toys and wall clocks use electricity from cells, which are also called batteries.

## The torch cell

A torch cell has two ends, called terminals. The end with a small metal cap is the positive terminal, marked with a plus sign. The flat end is the negative terminal, marked with a minus sign.

## The electric circuit

Take a cell, a small bulb and two wires. Join one wire from the positive terminal of the cell to the bulb, and another wire from the bulb back to the negative terminal. The bulb glows! The electricity has a complete path to flow from the cell, through the bulb and back to the cell. This complete path is called an electric circuit.

If one wire is removed, the path is broken and the bulb does not glow. A circuit with a complete path is called a closed circuit. A circuit with a broken path is called an open circuit.

A switch is used to open or close a circuit. When we switch on a fan, the circuit is closed. When we switch it off, the circuit is opened.

## Conductors and insulators

Make a circuit with a gap in it. Place different things in the gap one by one and see if the bulb glows.

- The bulb glows with an iron nail, a steel spoon, a copper wire and an aluminium foil. Materials that let electricity pass through them are called conductors. Most metals are good conductors. That is why electric wires are made of copper.
- The bulb does not glow with plastic, rubber, wood, glass or cloth. Materials that do not let electricity pass through them are called insulators. Electric wires are covered with plastic to keep us safe.

## Be safe with electricity

- Never touch a switch or a plug with wet hands.
- Never put your fingers or anything else into a plug socket.
- Stay away from fallen or broken electric wires.
- Never fly kites near electric wires.

## Save electricity

Switch off lights and fans when you leave a room. Use LED bulbs, which use less electricity than old bulbs.

## Let's think

1. What is an electric circuit?
2. Why are electric wires covered with plastic?
3. Write two ways to save electricity.

## New words

terminal - an end of a cell where wires are joined
conductor - a material that lets electricity pass through it